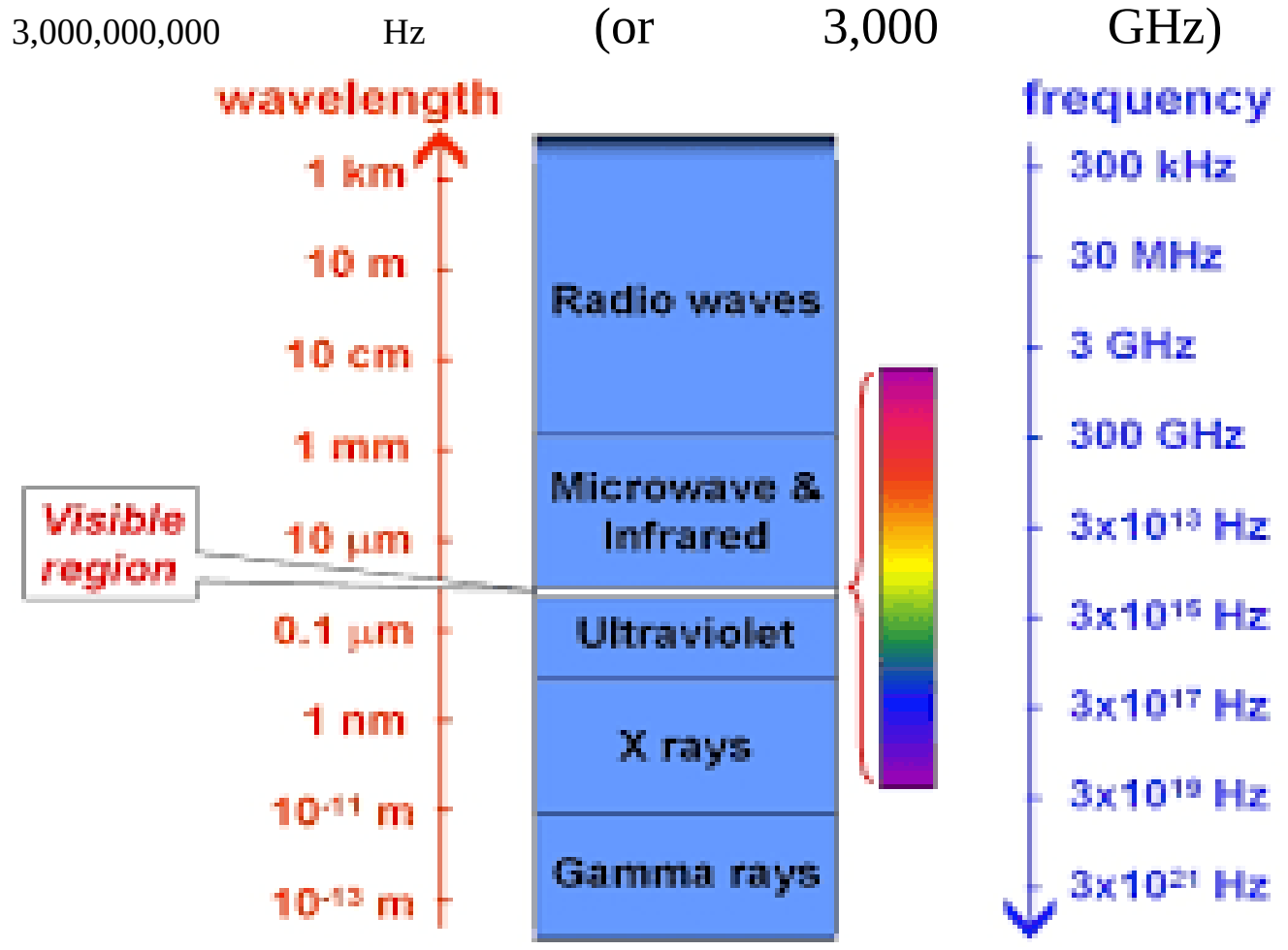


ELECTROMAGNETIC SPECTRUM AND SPECTRUM MANAGEMENT

The electromagnetic spectrum refers to the range of frequencies (the spectrum) of electromagnetic radiation. This electromagnetic radiation is made up of waves, of varying wavelengths, which carry energy. From shortest to longest wavelength, the full electromagnetic spectrum comprises gamma rays, x-rays, ultraviolet light, visible light, infrared light, microwaves and radiowaves. The radio spectrum refers to that part of the electromagnetic spectrum which comprises radio waves; these are waves with wavelengths of between 1mm and 10,000km. Waves are also measured according to their frequency. All waves travel at the same speed meaning the shorter the wavelength, the greater the number of waves that will pass through a given point during a particular period of time. The number of times a wave passes through a point in one second is called its frequency and is measured in Hertz (Hz). The radio spectrum is the part of the electromagnetic spectrum with frequencies from 3 Hz to



Radiation	Approximate wavelength range / m	Approximate frequency range / Hz
Radio	> 0.1	$< 3 \times 10^9$
Microwaves	$0.1 - 1 \times 10^{-3}$	$3 \times 10^9 - 3 \times 10^{11}$
Infra-red	$1 \times 10^{-3} - 7 \times 10^{-7}$	$3 \times 10^{11} - 4.3 \times 10^{14}$
Visible	$4 \times 10^{-7} - 7 \times 10^{-7}$	$7.5 \times 10^{14} - 4.3 \times 10^{16}$
Ultra-violet	$4 \times 10^{-7} - 1 \times 10^{-8}$	$7.5 \times 10^{14} - 3 \times 10^{16}$
X-rays	$1 \times 10^{-8} - 4 \times 10^{-10}$	$3 \times 10^{16} - 7.5 \times 10^{20}$
Gamma rays	$1 \times 10^{-10} - 1 \times 10^{-16}$	$3 \times 10^{18} - 3 \times 10^{24}$

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Practical Applications of Electromagnetic Waves

- The radio waves and microwaves discovered by Hertz paved the way for wireless television, radio and mobile communication.
- The visible light portion of the electromagnetic spectrum is the reason for all visual aids in daily life. This is the portion of the electromagnetic spectrum that helps us to see all objects, including colours.
- The X-rays discovered by Roentgen proved to be useful in medicine for detecting many ailments or deformities in bones.
- The high ultraviolet radiation has energies to ionize the atoms causing chemical reactions.
- The gamma rays discovered by Paul Villard are useful for ionisation purposes and nuclear medicine.

SPECTRUM REGULATORS

The International Telecommunication Union (ITU) is responsible for the global coordination of the radio frequency spectrum through the ITU Radio Communication Sector (ITU-R) Bureau. The ITU-R is charged with maintaining and extending international cooperation among all the Member States of ITU, for the improvement and rational use of radio communications of all kinds. In particular, it is responsible for the coordination and implementation of the Radio Regulations, which are annexed to the ITU Constitution and Convention. These Radio Regulations are revised by the ITU World Radio Communication Conferences (WRC), normally held every four years. Article 5 of the Radio Regulations contains the Table of Frequency allocations listing frequency bands allocated to the various services according to three ITU geographic regions.

The Communication authority of Kenya

The Communications Authority of Kenya (CA) The Communications Authority of Kenya (CA) is the regulatory authority for the communications sector in Kenya established through the Kenya Information and Communications Act 1998. The Authority is responsible for facilitating the development of the information and communications sectors including; broadcasting, multimedia, telecommunications, and postal services and electronic commerce.